Temporal Referent Integrity: An Evaluation Diagnostic for Resolution Timing in Tool-Using Agents

Anonymous submission

Abstract

Tool-using agents often act on entities after the environment changes. We study instructions that either commit to a still-valid entity before a refresh or defer a selector until afterward. Evaluating only stable cases or only one instruction type can miss whether a controller handles both cases. We introduce *temporal referent integrity* (TRI), a diagnostic built from matched Preserve/Reevaluate pairs. Pair members share the states, selector, action, and transition but require opposite targets when the selector winner changes. Pair accuracy (PairAcc) counts a pair only when both targets are correct. In a post-primary controlled replication across three model backends on 240 authored tasks spanning ten schemas, an author-designed Generic controller substituted the refreshed winner in 41/66, 30/70, and 50/69 cases where both probes exposed the correct initial binding; a Compile-then-act probe with an explicit decision record made no substitutions in those cases. A SQLite tool-loop test separately observed refreshed-winner writes after correct pre-refresh binding. Under equal calls and byte-identical base actor inputs, exposing a composite decision block improved authored PairAcc across four tested model backends. Effects on author-adapted public-benchmark pairs were model-dependent. TRI therefore provides a controlled diagnostic for selective re-resolution and its executed consequences; the present evaluation does not estimate native-task prevalence or unrestricted language transfer.

Introduction

Tool-using agents often select an entity, observe an updated environment, and act only afterward. During this interval, an update can change which entity satisfies the original selector. We study the resulting distinction between instructions that commit to the earlier entity and instructions that defer selection until after the update.

Consider an email agent. If it selects the highest-priority unread email and then refreshes the mailbox, a later request to reply to "it" refers to the selected email. If the instruction instead refreshes first and selects afterward, the same description is evaluated on the refreshed mailbox. Suppose email A wins before refresh and email B afterward. The first instruction targets A and the second targets B, although the states, selector, transition, and action are otherwise identical.

The same distinction appears in file-management dialogue. After an assistant reports a results folder, "put the experiment data in this folder" remains bound to that folder even if staging folders appear. By contrast, "use the most recently created results folder" defers selection until after the filesystem update.

An agent trace may retain the initial entity identifier (ID), the refreshed state, and the action schema while leaving this distinction implicit. Reapplying the selector can replace a committed referent; retaining the old ID can violate an instruction that deferred selection. Stable-only and one-sided evaluations can leave this difference unidentified. A Preserve-only test can reward Always-Lock, a Reevaluate-only test can reward Always-Reevaluate, and stable winners make the two policies observationally equivalent.

We introduce temporal referent integrity (TRI) as a controlled matched diagnostic. TRI represents an action reference as either *bound* to an entity or *deferred* as a selector. A Preserve/Reevaluate pair holds the states, selector, action, and transition fixed while changing the instruction phrasing that encodes commitment timing. When the selector winner changes, the two members require opposite targets. PairAcc counts a pair only when both targets are correct. This construction tests selective re-resolution under a shared transition.

Deterministic policy extremes establish what the paired endpoint distinguishes. Conditional substitution then tests whether a controller replaces a correct initial binding with the refreshed winner. A post-primary 240-task controlled replication spanning ten schemas measures this behavior with three model backends. A separate SQLite study follows the selected ID through a mutation issued by the model.

Controller probes test whether exposing a structured decision record changes this behavior after the initial selection is supplied. The primary comparison contrasts complete controller packages. An equal-call comparison then holds the base actor payload and call count fixed while exposing a composite decision block. PairAcc rises from 5/32 to 13/32 for Qwen and from 8/32 to 25/32 for GLM. Gains on pairs

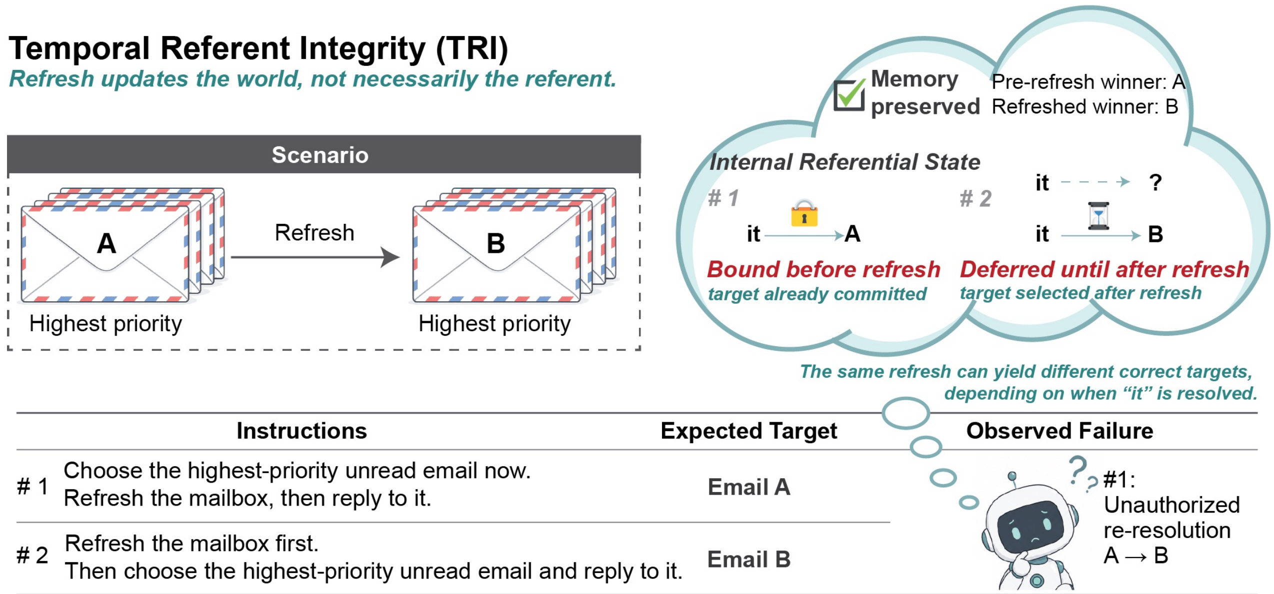

Figure 1: Temporal Referent Integrity under a shared transition. Email A wins before refresh and Email B afterward. Instruction 1 commits to A before refresh; Instruction 2 defers selection and targets B. Re-resolving Instruction 1 as B yields conditional target substitution despite the preserved state history.

adapted from public benchmarks are heterogeneous, and a strong event-order rule succeeds on the authored inventory but transfers poorly.

Our claim is diagnostic rather than architectural: the experiments test a controlled behavioral distinction, not the necessity of one record format, native-task prevalence, or open-language generalization.

The paper makes three contributions:

- We distinguish authorization from initial binding and post-commitment persistence, then show how crossed bound/deferred directions and opposite-gold PairAcc distinguish selective resolution within an exact-target three-policy class.
- A post-primary controlled replication localizes substitution after a correct binding; a separate SQLite tool-loop test shows that such substitution can reach wrong-entity writes. Initial-binding and fallback errors remain outside the TRI endpoint.
- Equal-call probes isolate the effect of exposing a composite timing decision across four backends; comparisons with full history, component controls, and a deterministic event-order rule show that the diagnostic does not depend on a unique architecture.

Related Work

Reference, memory, and changing state. Discourse semantics separates a referent from a description whose denotation can change with context (Kamp and Reyle 1993; Grosz, Joshi, and Weinstein 1995). Research on coreference and entity tracking studies how systems recover and maintain these identity links (Lee et al. 2017; Joshi et al. 2020; Tang et al. 2026). Agent-memory systems retain facts and interaction traces (Packer et al. 2023; Sumers et al. 2024), while temporal-agent benchmarks test evidence refresh, delayed intentions, and belief updates (Cheng et al. 2026; Liu and Gabriel 2026; Ji et al. 2026; Xu et al. 2026). TRI asks a narrower question after memory and state are available: which representation may be used to resolve the action reference at execution time?

Initial binding, persistence, and timing. Entity Binding studies initial tool-argument selection (Babu and Indukuri 2026b). Binding Drift assumes a primary referent has been committed, then tests persistence, reverification, and error propagation (Babu and Indukuri 2026a). TRI instead conditions on a correct initial binding, crosses bound and deferred authorization directions under the same transition, and jointly scores each opposite-gold Preserve/Reevaluate pair. It therefore evaluates whether re-resolution was authorized rather than proposing another persistence defense. Our Binding Drift author adaptation is neither an official reproduction nor an information-matched CTA baseline: it receives S_1 but neither S_0 nor the resolved old ID. Its complementary outcomes illustrate the interface boundary, not a performance advantage. Detailed counts are supplementary.

Behavioral and tool-agent evaluation. Behavioral tests and contrast sets expose capabilities that aggregate accuracy can miss (Ribeiro et al. 2020; Gardner et al. 2020). Stateful tool benchmarks add executable world changes, intermediate milestones, and target-level outcomes (Li et al. 2023; Liu et al. 2024; Lu et al. 2025; Trivedi et al. 2024; Yao et al. 2024; Sierra Research 2026; Yang et al. 2026). TRI changes resolution timing while holding the transition fixed. Joint scoring rejects two complementary policies, and the execution trace localizes errors after binding. Runtime ledgers and monitors (Uddin et al. 2026; Sohail and Haider 2026; Wang, Poskitt, and Sun 2025; Li et al. 2026) provide possible implementations. TRI evaluates the underlying resolution-time decision across such implementations.

Temporal Referent Integrity and Evaluation Identifiability

Let H be the instruction and interaction history, S_0 and S_1 the states before and after one refresh, r an action-target reference, and $q_r : S \to E \cup \{\perp\}$ its deterministic task resolver. Ties are excluded. We separate world evidence S, referential control state $C_H(r)$, and action validity $V_a(e, S)$. Immediately before refresh,

$$C_H(r) \in \{\mathrm{bound}(e_0), \mathrm{deferred}(q_r)\}, \quad e_0 = q_r(S_0) \in E. \tag{1}$$

$\mathrm{bound}(e_0)$ records an identity commitment. $\mathrm{deferred}(q_r)$ records a selector to evaluate after refresh. A world update can change the selector winner without changing a bound referent:

$$q_r(S_0) \neq q_r(S_1) \;\not\Rightarrow\; \mathrm{bound}(e_0) \mapsto \mathrm{bound}(q_r(S_1)). \tag{2}$$

In the actionable core, where the required target exists and satisfies the action preconditions, the correct target is

$$T^* = \begin{cases} e_0, & C_H(r) = \mathrm{bound}(e_0) \wedge V_a(e_0, S_1), \\ q_r(S_1), & C_H(r) = \mathrm{deferred}(q_r). \end{cases} \tag{3}$$

Here $e_1 = q_r(S_1) \in E$, and this actionable definition requires $V_a(T^*, S_1)$. Rows that violate this requirement, including those with $\neg V_a(e_0, S_1)$, form a separate fallback-policy slice. Reject, Clarify, and licensed reselection are execution choices beyond the referential identity estimand.

Observable substitution. We count conditional substitution only on Preserve rows with four observable properties: the initial target is correct; refresh completes; the selector winner changes; and the old target remains present and action-valid. A substitution occurs when the final target equals the refreshed winner. This endpoint separates post-binding replacement from initial grounding and validity errors. A state diff then tests whether the selected target receives the write.

Restricted identifiability observation. Within the exact-target policy class {Always-Lock, Always-Reevaluate, Selective}, evaluation requires at least one changed-winner Preserve row and one changed-winner Reevaluate row to distinguish Selective from both extremes. A two-row set is sufficient and minimal. Matching the rows on S_0, S_1, q, and a additionally controls task and transition content; it is not needed for the cardinality claim. The observation assumes deterministic policies, and the short argument is supplementary.

Stable tasks make lock and reevaluation observationally equivalent. A one-sided changed-winner test can reward one unconditional extreme. The crossed matched regime breaks both equivalences. For pair i, let Y_{Pi} and Y_{Ri} indicate exact correctness on its Preserve and Reevaluate members. Then

Evaluation regime	Lock	Reevaluate	Identifies?
Stable only	can pass	can pass	No
Changed Preserve only	can pass	fails	No
Changed Reevaluate only	fails	can pass	No
Matched changed PairAcc	zero	zero	Yes

Table 1: Policy identifiability. Stable or one-sided evaluation can favor unconditional policies; matched changed-winner PairAcc rejects both extremes within the three-policy class.

$$\begin{aligned} \mathrm{PairAcc} &= \frac{1}{N} \sum_i Y_{Pi} Y_{Ri}, \\ \max(0, A_P + A_R - 1) &\le \mathrm{PairAcc}, \\ \mathrm{PairAcc} &\le \min(A_P, A_R), \end{aligned} \tag{4}$$

where the quantities $A_P = N^{-1} \sum_i Y_{Pi}$ and $A_R = N^{-1} \sum_i Y_{Ri}$ are computed over the same N complete pairs. The marginals alone do not determine joint correctness. Crossed Preserve/Reevaluate coverage supplies the policy discrimination; matching controls task content, and PairAcc summarizes co-correctness within each pair. Table 1 gives the restricted policy comparison.

Aggregate certification gap. Suppose an N-row evaluation contains n disjoint complete actionable changed pairs and has aggregate end-to-end (E2E) accuracy A. Counting failed pairs gives the sharp certificate $\mathrm{PairAcc} \ge \max\{0, 1 - N(1-A)/n\}$. Hence $A = 1 - n/N$ remains compatible with zero PairAcc: one error in every critical pair can be diluted by non-pair rows. This is a worst-case guarantee, not evidence that aggregate selection fails in every candidate set.

Execution-risk decomposition. Let O denote a strict changed-winner opportunity, B a correct observable initial binding, U substitution to the refreshed winner, and X execution of the resulting unauthorized write. The chain rule gives

$$\begin{aligned} \Pr(O, B, U, X) &= \Pr(O) \Pr(B \mid O) \\ &\quad \times \Pr(U \mid O, B) \Pr(X \mid O, B, U). \end{aligned} \tag{5}$$

Conditional substitution estimates the third factor in the tested controller; it does not estimate opportunity prevalence, execution realization, or total tool-agent risk. Complete arguments and assumptions are supplementary.

We separately report exact target/final state, wrong-target writes, invalid attempts, and unnecessary rejections; always rejecting avoids writes but fails every actionable task.

Methods

Diagnostic Construction

Each domain specification defines an entity table, a natural-language selector, its ranking or Boolean criterion, an action, and explicit action preconditions. The generator first evaluates the selector on S_0 to obtain e_0, applies a controlled transition, and evaluates it on S_1 to obtain e_1. A language template determines whether the action target is e_0, e_1, or $\perp$. Paired tasks share the domain, both states, selector, action, and schema. Their instruction phrasing differs to encode whether the target is committed before refresh or selected afterward.

The crossed core combines Preserve/Reevaluate with an unchanged or changed selector winner. Stable controls expose unnecessary reactions to refresh. Changed-winner and name-collision cases create opposite-target pairs while retaining valid stable IDs. The Matched Timing Diagnostic contains 128 actionable rows and 32 Remove/Invalidate rows. The latter form a separate fallback-policy slice. The Cross-Schema Controlled Replication contains 40 state clusters across ten schemas. Each cluster contributes six rows: two reference modes crossed with Stable, Flip, and name-collision transitions. Flip and name-collision each form one changed-winner pair per cluster, yielding 80 changed pairs and the PairAcc denominator of 80 in Figure 4. Complete templates and transition taxonomies are supplementary.

Figure 2 summarizes the matched construction, shared probe interface, and distinct observable readouts; it is an evaluation workflow, not a controller architecture.

Measurements and Denominators

The frozen primary estimand is the Qwen all-row end-to-end (E2E) package difference between Generic and Lifecycle-Gated. PairAcc, actionable exact-target accuracy, and conditional substitution are the main diagnostic endpoints; initial binding, rejection, invalid attempts, and wrong-target writes remain separate. Results use intention-to-treat scoring and cluster-bootstrap intervals by state or workflow. Later intervals are descriptive. PairAcc measures co-correctness within frozen pairs, so it is not determined by the two marginals alone.

A deterministic executor checks whether each selected target becomes the written target in the refreshed state. A separate 40-task SQLite tool loop records model-driven final states, wrong writes, invalid attempts, and collateral changes.

Controller Probes

Generic Structured Ledger (Generic) retains the instruction, selected ID, entity snapshot, selector, action, and preconditions. It does not encode the reference mode (bound or deferred). Compile-then-act (CTA) emits a pre-refresh record containing the reference mode, selector, and bound ID. Lifecycle records add validity and fallback fields, while actor-mediated and gated variants test soft and hard use of that record. These large language model (LLM) controllers are operational probes for the same timing decision. The typed record and gate branches are supplementary. Generic exposes both the old selected entity and the current selector result but does not prescribe a timing policy.

Always-Lock and Always-Reevaluate provide deterministic policy extremes. The former retains the S_0 target when valid; the latter applies q to S_1. A timing reminder tests whether ordinary history can expose the distinction without a structured record. Rule* is a deterministic event-order parser written after inspection of earlier errors and is reported as post-hoc.

Controllers receive the instruction's natural-language selector. Gold targets, normalized selector fields, and pre- and post-refresh winner IDs are withheld. The primary Qwen comparison is a package contrast with unequal calls because Lifecycle-Gated skips an actor call on valid Preserve rows. Later equal-call experiments compare History-only and Decision-visible actors using the same base payloads, states, tool schemas, and call count. The visible block jointly contains the predicted reference mode, bound ID, and selector restatement; Decision-enforced applies it offline. The same interface is used for authored, rewritten, and public-benchmark-adapted tasks. Prompts, payloads, and exact scoring rules are supplementary. The PDF supplement gives the verbatim interface; prompts, payloads, raw outputs, and scoring code are in the anonymous Code and Data Supplement.

Construct and transfer protocols. One adult volunteer rewrote 50 sampled scalar instructions. Three other adults independently labeled 100 randomized original/rewrite items using opaque IDs without condition metadata or gold. Referent identity is analyzed separately from Reject, Clarify, or reselection. Participants gave informed consent; released responses are de-identified. The study was reviewed by the authors' institution and determined to be exempt. The transfer audit uses 30 author-constructed pairs over states and tool schemas from three public resources, with matched interfaces and call counts.

Evidence Status

The Qwen package comparison is primary/frozen, with a later GLM replication. The ten-schema controller audit, equal-call contrasts, human agreement audits, and external audits are post-primary; Rule* is post-hoc. The supplement gives the complete chronology, denominators, intervals, and failure accounting. Paper-facing backends are Qwen3.5-122B-A10B (Qwen), GLM-5.1 (GLM), DeepSeek-V4-Pro (DeepSeek), and, in four-model matched controls, MiniMax-M2.5 (MiniMax). Each frozen experiment records its exact model set and temperature-zero settings. The artifact contains frozen inventories, prompts, outputs, reports, hashes, and the error taxonomy.

These evidence families answer different questions and are not pooled. Deterministic controls establish policy discrimination; authored runs spanning ten schemas locate substitution after correct binding; the SQLite subset tests execution; equal-call contrasts test the composite decision block; and external adaptations and retrieval audits probe transfer and observed coverage.

Results

The results ask whether the paired design identifies selective behavior, whether substitution reaches execution, whether decision visibility changes equal-call outcomes, and how far the construct transfers.

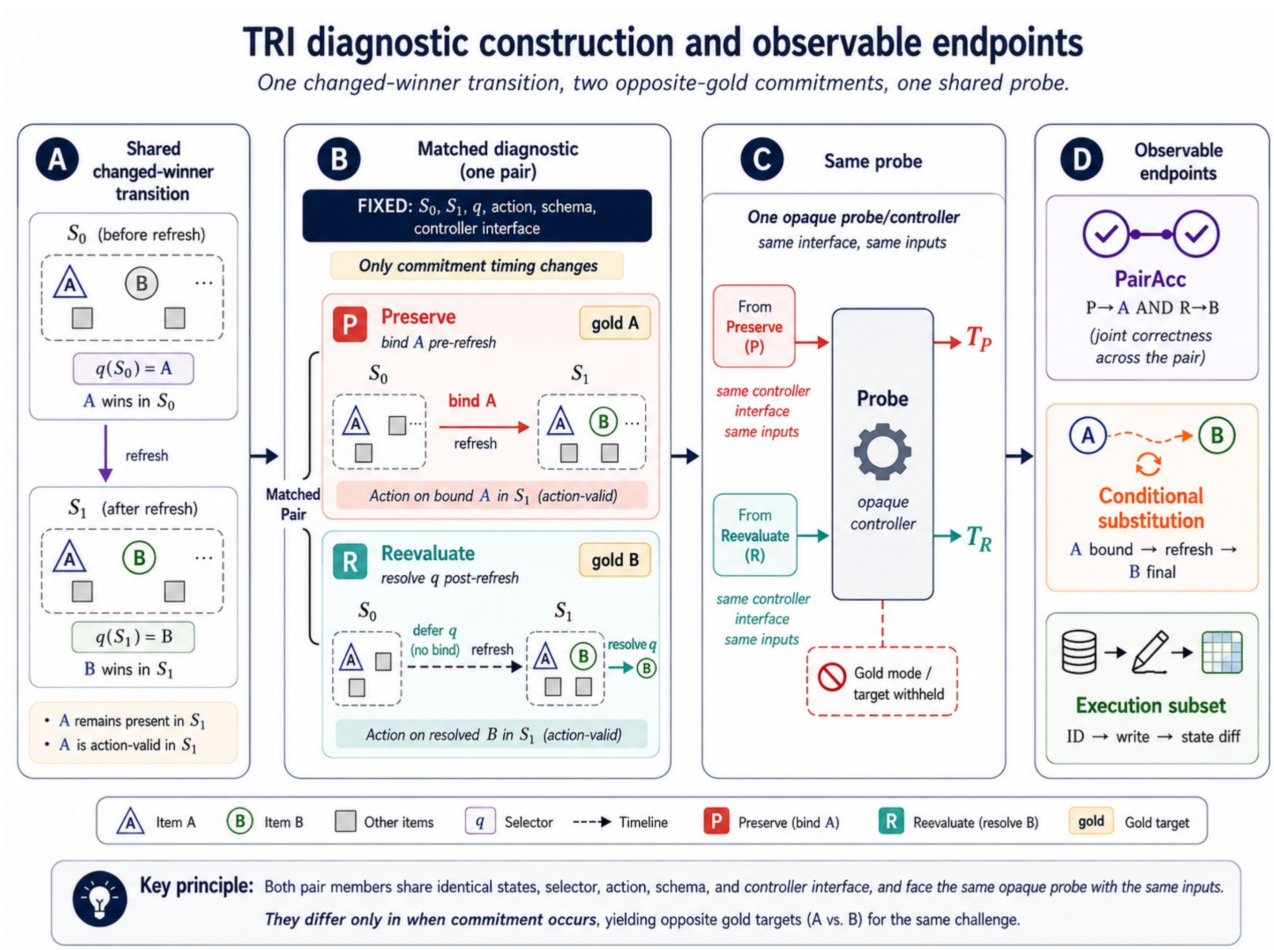

Figure 2: TRI diagnostic construction and observable endpoints. A changed-winner pair holds S_0, S_1, selector q, action, schema, and controller interface. A wins in S_0; B wins in S_1, while A remains present and action-valid. Preserve binds A before refresh, whereas Reevaluate defers q and targets B afterward. PairAcc requires both outputs to be correct, conditional substitution isolates $A \to B$ after a correct binding, and the execution subset records which entity receives the write. Gold resolution modes and targets are withheld from the probe.

RQ1: Does Pairing Identify Selective Resolution?

Policy discrimination. On the 160-row Matched Timing Diagnostic, Always-Lock and Always-Reevaluate each score 60.0% in aggregate, but succeed on complementary instruction types and score 0/32 changed-winner PairAcc (Figure 3). Across five tested controller candidate sets, aggregate E2E and PairAcc choose the same policy; there is no ranking reversal. Rule* is post-hoc, and its strong score limits algorithmic novelty.

RQ2: Does Substitution Reach Execution?

Primary package comparison. The primary comparison changes the complete controller package and uses different numbers of model calls. Across 160 tasks, Qwen E2E changes from 103/160 for Generic to 157/160 for Lifecycle-Gated (+33.8 points; language-template-cluster 95% confidence interval (CI) [18.1, 50.0]); the later GLM replication changes from 115/160 to 160/160 (+28.1 points; 95% CI [18.1, 38.1]). On 128 actionable tasks, Qwen Lifecycle-Gated reaches 125/128 versus Generic 95/128; GLM CTA reaches 128/128 versus Generic 93/128. Fallback cases are supplementary. This is a package contrast, not the effect of one timing field.

Conditional target substitution. We condition on a correct initial ID, completed refresh, distinct new winner, and surviving valid old target, excluding initial-binding, tool-order, and validity failures. In the 240-task, ten-schema replication, Generic substitutes on 41/66, 30/70, and 50/69 rows eligible under both controllers; CTA substitutes on none (Figure 4). The pattern appears for Qwen, GLM, and DeepSeek under the same author-designed controller.

Executed target consequences. In the 40-task SQLite tool-loop test, Generic reaches 27/40 correct final states with 13 wrong-target writes for Qwen, and 26/40 with eight for GLM. Among strict opportunities, refreshed-winner writes

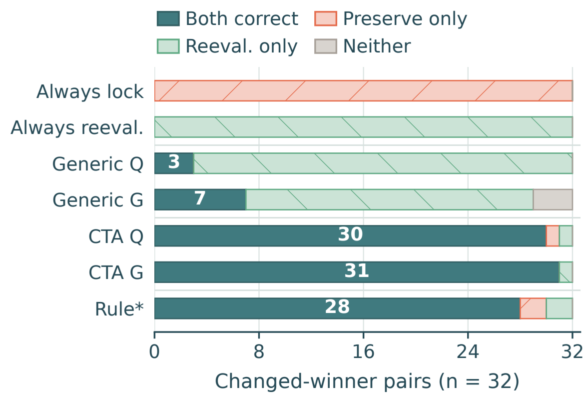

Figure 3: Joint outcomes on 32 actionable changed-winner pairs. Each bar partitions pairs into both correct, Reevaluate-only, Preserve-only, and neither-correct outcomes. Always-Lock and Always-Reevaluate succeed on complementary members but never complete a pair; Rule* is post-hoc.

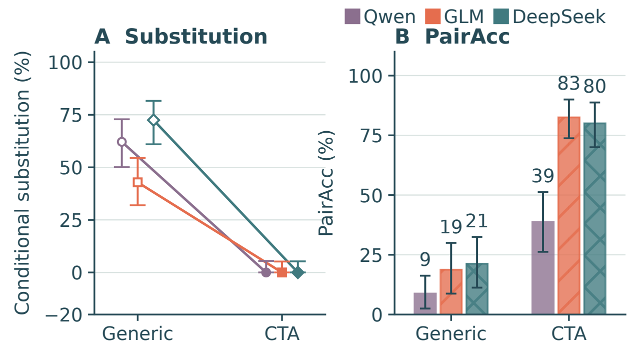

Figure 4: Ten-schema outcomes after correct initial binding. A: Generic versus CTA substitution on rows eligible under both controllers (Wilson 95% CIs); open Generic and filled CTA endpoints are linked within model, and zero CTA estimates lie at the lower bound. B: changed-winner PairAcc over 80 pairs (cluster-bootstrap 95% CIs); labels are rounded percentages. Counts (Qwen/GLM/DeepSeek) are Generic 7/15/17 of 80 and CTA 31/66/64 of 80. CTA makes no common-eligible substitutions; neither endpoint is a safety rate.

occur in 8/8 and 6/8 Changed cases, versus 0/4 Stable controls for each model (Figure 5). Remaining wrong writes are confined to fallback cases; GLM also produces six unneeded rejections. These trajectories locate the failure between binding and action.

A fixed-executor replay turns every Generic substitution in the 240-task replication into a wrong-target write. This target-to-write consistency check is supplementary.

RQ3: Does Decision Visibility Change Outcomes under Equal Calls?

Two post-primary equal-call audits test different ways of exposing timing information on separate frozen inventories (Figure 6). Convention-told effects are model-conditional, with Qwen and DeepSeek intervals including zero, whereas Decision-visible yields positive changed-winner PairAcc effects with all four intervals above zero. The inventories are not pooled; Convention-told and decision-block runs contain 1 and 0 incomplete ITT rows, respectively.

Across nine audited matched-call inventories, both actors receive the same initial ID; the visible selector and initial-ID values are exact repetitions in 760/760 audited records. This excludes new field values, but not repetition, salience, or structured-framing effects. Compiler-output strata are supplementary and descriptive.

A frozen three-model, eight-cell decomposition further separates placebo structure, selector and ID repetition, mode, their combinations, and an explicit follow directive. No predeclared field-level contrast met the cross-model promotion gate; component effects were model-dependent, so the supported intervention remains composite rather than attributable to one field.

RQ4: How Far Does the Diagnostic Transfer?

Construct scope. In the earlier three-annotator convenience sample, majority–gold agreement is 98.0% on 50 dynamic items and 86.7% on 30 anchored actionable items, but 55.0% on 20 action-invalid REJECT items. We therefore center actionable referent identity and treat fallback policy separately.

Human rewrites. On the 48 rewrites with determinate majority labels, CTA reaches 89.6% for Qwen and 93.8% for GLM. Equal-call actionable E2E is 30/40 for both Qwen conditions and rises from 31/40 to 39/40 for GLM. Only three changed pairs are complete, so PairAcc is descriptive. These reformulations test wording robustness within one volunteer's rewrites rather than population-level language transfer.

Pairs adapted from public benchmarks. The 30-pair test uses states and tool schemas from STATE-Bench, AgentDojo, and ToolSandbox, with author-constructed interventions. A temperature-zero repeat remains model-conditional: changed PairAcc effects are 16.7, 23.3, and 6.7 points for Qwen, GLM, and DeepSeek; a first MiniMax pass is 3.3 points. Repeats measure endpoint stability, not new samples, native behavior, or prevalence (full results in the supplement).

Representation and rule boundaries. CTA, Lifecycle, and a full-history timing reminder improve some authored conditions, but no representation dominates across models and datasets. Against final-step-aware full history on the same 240 rows, CTA E2E differs by $+1.2$ $[-6.7, 9.2]$ points for Qwen, $+13.3$ $[8.8, 17.9]$ for GLM, and $+15.4$ $[9.2, 21.7]$ for DeepSeek; the Qwen interval includes zero. Rule*, written after error inspection, reaches 92.5% on Matched Timing Diagnostic, 96.0% on rewrites, and 91.7% on Cross-Schema Controlled Replication, showing usable event-order cues in the authored inventory. After freezing, it reaches only 15/60 exact targets and 2/30 PairAcc on the public-benchmark

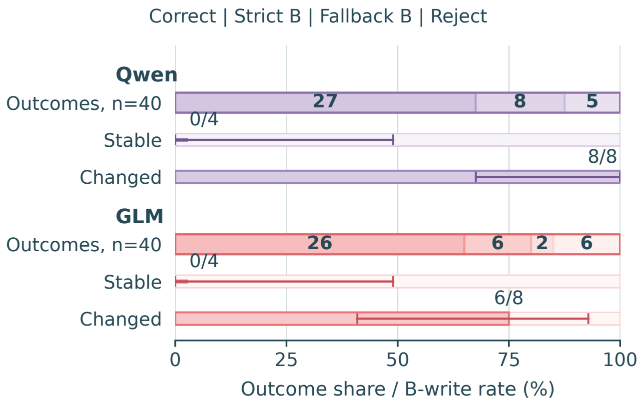

Figure 5: Complete 40-task SQLite outcomes and strict refreshed-winner writes for Generic. Outcome strips partition each model's trajectories into correct final states, strict B writes, fallback B writes, and unneeded rejections (left to right). Stable keeps A ranked first; Changed makes B first while A remains valid. Strict B-write rates use Wilson 95% CIs; exact fractions are 0/4 Stable for both models and 8/8 Qwen, 6/8 GLM in Changed. This controlled test is not a prevalence estimate.

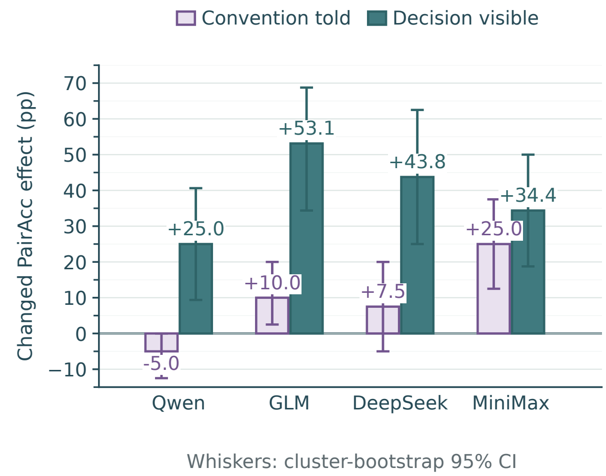

Figure 6: Post-primary equal-call changed-winner PairAcc effects. Left lavender bars: Convention-told versus Plain history (40 changed pairs); right teal bars: Decision-visible versus History-only (32 actionable changed pairs). Whiskers are cluster-bootstrap 95% CIs. Inventories are separate and unpooled; side-by-side placement is descriptive.

adaptations, where unfamiliar rankings introduce parsing and grounding errors.

A post-primary oracle decomposition identifies a separate grounding boundary. On an 80-row unseen-schema stress set, Qwen mode is 80/80 but the Preserve bound ID is 21/40. Replacing the ID raises gated accuracy from 66/80 to 78/80; replacing mode does not. In this Qwen 80-row stress set, the remaining loss is concentrated in selector grounding.

Coverage in native tasks. A frozen 96-task ToolSandbox-style extension finds zero substitutions in four Qwen/GLM controller conditions over 64–87 eligible rows each. Across 20 STATE-Bench and AgentDojo clusters (Microsoft Research 2026; Debenedetti et al. 2024), only Qwen ordinary history on AgentDojo shows substitutions: 2/7 changed rows versus 0/7 matched Stable rows. The other combinations of repository and model are zero, and no method consistently improves execution accuracy. These are conditional endpoints, not overall safety rates.

An author-built retrieval audit finds no case meeting all eligibility conditions in six public benchmarks. Synthetic controls verify the checklist, but natural retrieval recall is uncalibrated.

Composition and repeatability. Composition tests are mixed: a two-refresh Qwen test favors Generic over scalar Lifecycle (32/40 vs. 28/40), while held-out role indexing improves 35/40 to 39/40. Across three temperature-zero passes, CTA remains above Generic and records no substitutions. Full results are supplementary.

Discussion

TRI places the action reference on opposite sides of a shared transition. PairAcc tests joint correctness, conditional substitution localizes post-binding failures, and SQLite traces identify the written entity. The candidate sets reveal complementary success patterns without changing the aggregate-selected controller; the diagnostic need not reverse every ranking.

Matched-interface block exposure improves authored PairAcc but identifies neither a unique architecture nor a field-specific effect; enforcement can preserve a wrong compiled decision. The eight-cell decomposition likewise finds no component contrast that promotes across models. Rule*'s authored strength and external failure separate event-order extraction from selector grounding. Transfer remains model-dependent: only GLM has a clear external-adaptation E2E gain, the low-intervention extension is null, and the retrieval audit finds no native eligible case.

Limitations

Evaluation centers controlled single-refresh tasks and authored timing contrasts. Volunteer rewrites add limited wording variation, while public-suite retrieval has uncalibrated recall and does not estimate native prevalence. Agreement is higher for referent identity than for fallback policy; ambiguity, invalid targets, multiple refreshes, and referent composition require richer policies. The primary package contrast uses unequal calls, and matched-call tests isolate only composite decision visibility after the initial ID. Provider weights lack immutable revision IDs.

Conclusion

TRI is a matched test of referential commitment across refreshes. Within the tested exact-target class, changed-winner PairAcc separates selective from unconditional policies, and execution probes connect target replacement to writes.

Complete-block exposure improves authored PairAcc at four matched hosted endpoints, with model-dependent transfer. Together, the results establish a controlled diagnostic for comparing resolution-time decisions and their execution consequences without privileging a unique record format.

References

Babu, R. S.; and Indukuri, S. 2026a. Binding Drift in Multi-Step Tool-Augmented Agents. Concurrent preprint, arXiv:2607.18316.

Babu, R. S.; and Indukuri, S. 2026b. Entity Binding Failures in Tool-Augmented Agents. arXiv:2606.30531.

Cheng, Y.; Moakhar, A. S.; Fan, C.; Hosseini, P.; Faghih, K.; Sodagar, Z.; Wang, W.; and Feizi, S. 2026. Your LLM Agents Are Temporally Blind: The Misalignment Between Tool Use Decisions and Human Time Perception. Findings of ACL 2026, arXiv:2510.23853.

Debenedetti, E.; Zhang, J.; Balunović, M.; Beurer-Kellner, L.; Fischer, M.; and Tramèr, F. 2024. AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents. In *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*.

Gardner, M.; Artzi, Y.; Basmova, V.; Berant, J.; Bogin, B.; Chen, S.; Dasigi, P.; Dua, D.; et al. 2020. Evaluating Models' Local Decision Boundaries via Contrast Sets. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, 1307–1323.

Grosz, B. J.; Joshi, A. K.; and Weinstein, S. 1995. Centering: A Framework for Modeling the Local Coherence of Discourse. *Computational Linguistics*, 21(2): 203–225.

Ji, H.; Xiong, K.; Han, S.; Xia, P.; Qiu, S.; Zhou, Y.; Liu, J.; Li, J.; Li, B.; Zheng, Z.; Xie, C.; and Yao, H. 2026. ClawArena: Benchmarking AI Agents in Evolving Information Environments. arXiv:2604.04202.

Joshi, M.; Chen, D.; Liu, Y.; Weld, D. S.; Zettlemoyer, L.; and Levy, O. 2020. SpanBERT: Improving Pre-training by Representing and Predicting Spans. *Transactions of the Association for Computational Linguistics*, 8: 64–77.

Kamp, H.; and Reyle, U. 1993. *From Discourse to Logic: Introduction to Model-Theoretic Semantics of Natural Language, Formal Logic and Discourse Representation Theory*. Kluwer Academic Publishers.

Lee, K.; He, L.; Lewis, M.; and Zettlemoyer, L. 2017. End-to-end Neural Coreference Resolution. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, 188–197.

Li, M.; Zhao, Y.; Yu, B.; Song, F.; Li, H.; Yu, H.; Li, Z.; Huang, F.; and Li, Y. 2023. API-Bank: A Comprehensive Benchmark for Tool-Augmented LLMs. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 3102–3116.

Li, Y.; Chen, Y.; Wen, H.; Zhang, B.; Liu, H.; Wang, P.; Feng, Y.; and Tian, Y. 2026. VIGIL: Runtime Enforcement of Behavioral Specifications in AI Agent Skills. arXiv:2606.26524.

Liu, G.; and Gabriel, S. 2026. PM-Bench: Evaluating Prospective Memory in LLM Agents. In *Conference on Language Modeling*. ArXiv:2607.12385.

Liu, X.; Yu, H.; Zhang, H.; Xu, Y.; Lei, X.; Lai, H.; Gu, Y.; Ding, H.; Men, K.; et al. 2024. AgentBench: Evaluating LLMs as Agents. In *International Conference on Learning Representations*.

Lu, J.; Holleis, T.; Zhang, Y.; Aumayer, B.; Nan, F.; Bai, H.; Ma, S.; Ma, S.; Li, M.; Yin, G.; Wang, Z.; and Pang, R. 2025. ToolSandbox: A Stateful, Conversational, Interactive Evaluation Benchmark for LLM Tool Use Capabilities. In *Findings of the Association for Computational Linguistics: NAACL 2025*, 1160–1183.

Microsoft Research. 2026. STATE-Bench: Benchmark for Enterprise Workflows. GitHub software release. Pinned commit 0962c71.

Packer, C.; Wooders, S.; Lin, K.; Fang, V.; Patil, S. G.; Stoica, I.; and Gonzalez, J. E. 2023. MemGPT: Towards LLMs as Operating Systems. arXiv:2310.08560.

Ribeiro, M. T.; Wu, T.; Guestrin, C.; and Singh, S. 2020. Beyond Accuracy: Behavioral Testing of NLP Models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4902–4912.

Sierra Research. 2026. τ^3-Bench 1.0.0: Voice, Knowledge, and Task Quality. GitHub software release.

Sohail, S.; and Haider, G. 2026. Bounded Autonomy for Enterprise AI: Typed Action Contracts and Consumer-Side Execution. arXiv:2604.14723.

Sumers, T. R.; Yao, S.; Narasimhan, K.; and Griffiths, T. L. 2024. Cognitive Architectures for Language Agents. *Transactions on Machine Learning Research*.

Tang, Z.; Zhao, Q.; Franco, G.; Wijaya, D.; Mueller, A.; Schuster, S.; and Kim, N. 2026. Do Language Models Track Entities Across State Changes? In *International Conference on Machine Learning*. ArXiv:2605.30233.

Trivedi, H.; Khot, T.; Hartmann, M.; Manku, R.; Li, V. D.; Gupta, S.; Sabharwal, A.; and Balasubramanian, N. 2024. AppWorld: A Controllable World of Apps and People for Benchmarking Interactive Coding Agents. arXiv:2407.18901.

Uddin, M. N.; Saeidi, A.; Blanco, E.; and Baral, C. 2026. Ledger-Agent: Structured State for Policy-Adherent Tool-Calling Agents. arXiv:2606.20529.

Wang, H.; Poskitt, C. M.; and Sun, J. 2025. AgentSpec: Customizable Runtime Enforcement for Safe and Reliable LLM Agents. arXiv:2503.18666.

Xu, J.; Li, Q.; Wu, J.; Lan, Y.; Li, S.; Zhou, H.; Jiang, B.; Wang, L.; Wang, J.; Luu, A. T.; Xiong, C.; Park, H. W.; Hooi, B.; and Hu, Z. 2026. EvoArena: Tracking Memory Evolution for Robust LLM Agents in Dynamic Environments. arXiv:2606.13681.

Yang, L.; Wang, Z.; Tang, X.; Zhou, S.; Chen, D.; Jiang, W.; and Li, Y. 2026. ProBench: Benchmarking GUI Agents with Accurate Process Information. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 27547–27555.

Yao, S.; Shinn, N.; Razavi, P.; and Narasimhan, K. 2024. Tau-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains. arXiv:2406.12045.